

A solution to Haagerup’s problem on normal weights

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Abstract

We will introduce three open problems related to normal weights posed by Haagerup fifty years ago in his master’s thesis. Among these, we especially discuss a solution to the last problem and its meaning. This will provide a complete set of the positive Hahn-Banach separation theorems on operator algebras and shed light on why the weak* topologies on the duals of C^* -algebras are relatively difficult to work with. (arXiv:2501.16832.)

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1 Weights

U. Haagerup, *Normal weights on W^* -algebras*, J. Funct. Anal. (1975)

1.1 Normality

We will always denote a C^* -algebra and a von Neumann algebra by A and M respectively.

Definition. A *weight* on A is a homogeneous additive functional $\varphi : A^+ \rightarrow [0, \infty]$. A *subadditive weight* on A is a homogeneous subadditive functional $\varphi : A^+ \rightarrow [0, \infty]$.

Definition. For a weight φ on A , we say it is

- (i) *faithful* if $\varphi(a) = 0$ implies $a = 0$ for $a \in A^+$,
- (ii) *densely defined* if $\varphi^{-1}([0, \infty))$ is norm dense in A^+ ,
- (iii) *lower semi-continuous* if $\varphi^{-1}([0, 1])$ is norm closed in A^+ .

For a weight φ on M , we say it is

- (ii') *semi-finite* if $\varphi^{-1}([0, \infty))$ is σ -weakly dense in M^+ ,
- (iii') *normal* if ...?

Theorem (Haagerup, 1975). *For a weight φ on a von Neumann algebra, the followings are all equivalent.*

- (1) φ is completely additive for positive elements.
- (2) φ preserves directed suprema.
- (3) φ is σ -weakly lower semi-continuous.
- (4) φ is given by the pointwise supremum of normal positive linear functionals.

Remark. Each step in decreasing direction is easy. Haagerup proved (1) $\Rightarrow$ (3) and (3) $\Rightarrow$ (4).

Remark. Every von Neumann algebra admits a faithful semi-finite normal weight. There is a C^* -algebra that does not admit a faithful densely defined lower semi-continuous weight.

1.2 Semi-cyclic representations

Definition. A *semi-cyclic representation* of a C^* -algebra A is a representation $\pi : A \rightarrow B(H)$ together with a partially defined left A -linear operator $\Lambda : A \rightarrow H$ with dense range.

Proposition. *There is a one-to-one correspondence between weights on M and the unitary equivalence classes of semi-cyclic representations.*

Proposition. *For a weight φ on M and its associated semi-cyclic representation (π, Λ) of M , φ is faithful if and only if Λ is injective, φ is semi-finite if and only if Λ is densely defined, and φ is normal if and only if π is normal and Λ is σ -weakly closed.*

2 Haagerup's three problems

2.1 Problem 1.10

$$\left\{ \begin{array}{l} \text{For a subadditive weight on a von Neumann algebra,} \\ \text{is it } \sigma\text{-weakly lower semi-continuous if it preserves directed suprema?} \end{array} \right\}$$

Haagerup provided a proof of (1) $\Rightarrow$ (3) directly without an intermediate step (2). The proof is divided into two parts. First, it is proved for σ -finite von Neumann algebras, and extended to general von Neumann algebras. In a σ -finite von Neumann algebra, the σ -strong topology on the unit ball is metrizable. The additivity of a weight is used in the proof for the case of σ -finite von Neumann algebras, so it is enough to solve in the σ -finite case for this problem.

2.2 Problem 1.11

$$\left\{ \begin{array}{l} \text{For a weight on a von Neumann algebra,} \\ \text{is it normal if it is normal on every commutative von Neumann subalgebra?} \end{array} \right\}$$

Theorem (Dixmier, 1953). *For a positive linear functional ω on M , the followings are all equivalent.*

- (0) ω is completely additive for orthogonal projections.
- (1) ω is completely additive for positive elements.
- (2) ω preserves directed suprema.
- (3) ω is σ -weakly continuous.

Corollary. *A (positive) linear functional on M is normal if it is normal on every commutative von Neumann subalgebra.*

Remark. This corollary is used to prove some equivalent characterizations for weak compactness in M_* .

Remark. (0) $\Rightarrow$ (1) is false for weights.

2.3 Problem 2.7

$$\left\{ \text{Does the positive bipolar theorem hold for dual } C^*\text{-algebras?} \right\}$$

Definition. Let (E, E^*) be a dual pair of (directed partially) ordered real vector spaces such that E^+ and E^{*+} are mutually dual cones, i.e. $E^{*+} = \{x^* \in E^* : x^*(x) \geq 0 \text{ for } x \in E\}$. For a subset F of E^+ , we say it is *hereditary* if $0 \leq x \leq y \in F$ implies $x \in F$, and its *positive polar* is defined by the positive part of the real polar

$$F^{r+} := \{x^* \in E^{*+} : \sup_{x \in F} x^*(x) \leq 1\}.$$

Theorem (Haagerup 1975). *Consider the ordered real dual pairs (M^{sa}, M_*^{sa}) and (A^{sa}, A^{sa*}) .*

- (a) *If F is a σ -weakly closed convex hereditary subset of M^+ , then $F = F^{r+r+}$.*
- (b) *If F_* is a norm closed convex hereditary subset of M_*^+ , then $F_* = F_*^{r+r+}$.*
- (c) *If F is a norm closed convex hereditary subset of A^+ , then $F = F^{r+r+}$.*

Theorem (C. 2025).

- (d) *If F^* is weakly* closed convex hereditary subset of A^{*+} , then $F^* = (F^*)^{r+r+}$.*

Remark. They can be written in the form of Hahn-Banach separation theorems.

2.4 Corollaries of positive bipolar theorems

Corollary (Haagerup, 1975). *For a weight φ on a von Neumann algebra, the followings are all equivalent.*

(3) φ is σ -lower semi-continuous.

(4) φ is given by the pointwise supremum of normal positive linear functionals.

Proof. Let φ be a σ -weakly lower semi-continuous subadditive weight on a von Neumann algebra M . Define

$$F := \{x \in M^+ : \varphi(x) \leq 1\}, \quad F_* := \{\omega \in M_*^+ : \omega \leq \varphi\}.$$

Note that

$$F_* = F^{r+} := \{\omega \in M_*^+ : \sup_{x \in F} \omega(x) \leq 1\}$$

Then, the statement

$$\varphi(x) = \sup_{\omega \in F_*} \omega(x), \quad x \in M^+,$$

is equivalent to

$$F = F_*^{r+} := \{x \in M^+ : \sup_{\omega \in F_*} \omega(x) \leq 1\}.$$

Since F is σ -weakly closed by the σ -weak lower semi-continuity of φ , and since F is clearly convex and hereditary by definition of subadditive weights, we have $F = F^{r+r+}$ by the part (a) of the previous theorem. $\square$

Corollary (Combes, 1968). *For a weight φ on a C^* -algebra, the followings are all equivalent.*

(3) φ is lower semi-continuous.

(4) φ is given by the pointwise supremum of positive linear functionals.

Proof. Same as above but by (c) instead of (a). $\square$

Remark. (c) follows from the direct application of (a), but it can be also shown by purely C^* -algebraic methods with unitization, and this is what Combes essentially did in 1968. (probably?)

Corollary. *There are one-to-one correspondences*

$$\left\{ \begin{array}{c} \text{normal} \\ \text{subadditive} \\ \text{weights on } M \end{array} \right\} \leftrightarrow \left\{ \begin{array}{c} \sigma\text{-weakly closed} \\ \text{convex hereditary} \\ \text{subsets of } M^+ \end{array} \right\} \leftrightarrow \left\{ \begin{array}{c} \text{norm closed} \\ \text{convex hereditary} \\ \text{subsets of } M_*^+ \end{array} \right\}$$

and

$$\left\{ \begin{array}{c} \text{lower semi-continuous} \\ \text{subadditive} \\ \text{weights on } A \end{array} \right\} \leftrightarrow \left\{ \begin{array}{c} \text{norm closed} \\ \text{convex hereditary} \\ \text{subsets of } A^+ \end{array} \right\} \leftrightarrow \left\{ \begin{array}{c} \text{weakly}^* \text{ closed} \\ \text{convex hereditary} \\ \text{subsets of } A^{*+} \end{array} \right\}$$

Remark. For a subadditive weight φ on A , it is a weight if and only if

$$G^* := \{\omega \in A^{*+} : \exists \varepsilon > 0, (1 + \varepsilon)\omega \leq \varphi\}$$

is directed. Similarly for von Neumann algebras.

3 Proofs

3.1 Proof of (a)

Definition (Suppression by the one-parameter family of functional calculi). For $\delta > 0$, we define a function $f_\delta : (-\delta^{-1}, \infty) \rightarrow \mathbb{R}$ such that

$$f_\delta(t) := t(1 + \delta t)^{-1}, \quad t > -\delta^{-1}.$$

Then, f_δ is operator monotone, σ -strongly continuous, and enjoys a semi-group property, etc.

Proof sketch of (a) by Haagerup. Since the positive polar and the positive bipolar can be written as

$$\begin{aligned} F^{r+} &= F^r \cap M_*^+ = F^r \cap (-M^+)^r = (F \cup -M^+)^r = (F - M^+)^r, \\ F^{r+r+} &= (F - M^+)^{rr+} = (\overline{F - M^+})^+ \end{aligned}$$

by the usual real bipolar theorem and $F = (F - M^+)^+ \subset (\overline{F - M^+})^+$, we need to show $(\overline{F - M^+})^+ \subset F$.

Heuristics: Let $x \in (\overline{F - M^+})^+$ with nets x_i and y_i such that $x_i \rightarrow x$ σ -weakly in M and $x_i \leq y_i \in F$ for all i . Observe that if x_i were bounded by $r > 0$, then assuming $x_i \rightarrow x$ σ -strongly and $f_\delta(y_i) \rightarrow y_\delta$ σ -weakly for each $0 < \delta < r^{-1}$, we get

$$x_i \leq y_i \in F, \quad f_\delta(x_i) \leq f_\delta(y_i) \in F, \quad 0 \leq f_\delta(x) \leq y_\delta \in F, \quad f_\delta(x) \in F, \quad x \in F.$$

However, x_i may not be bounded, so we want to find a proof that uses the Krein-Šmulian theorem.

Solution: Define

$$G := \{x \in M^{sa} : \text{for any sufficiently small } \delta > 0, f_\delta(x) \in F - M^+\}.$$

Since $F - M^+ \subset G$ and $G^+ \subset F$, we are done if we prove G is σ -weakly closed in M . Fix $r > 0$ arbitrarily and let $M_r := \{x \in M : \|x\| \leq r\}$. The proof of the σ -weak closedness is divided into the two steps: $G \cap M_r$ is σ -strongly closed, $G \cap M_r$ is convex. Now we can deduce that, because G is convex and $G \cap M_r$ is σ -weakly closed, G is σ -weakly closed by the Krein-Šmulian theorem. $\square$

3.2 Proof of (b)

Definition (Bounded commutant Radon-Nikodym derivatives). Let $\psi \in M_*^+$ and let $\pi : M \rightarrow B(H)$ be associated cyclic representation to ψ with the canonical cyclic vector $\Omega \in H$. Then, we have a positive bounded linear map $\theta : \pi(M)' \rightarrow M_*$ defined such that

$$\theta(h)(x) := \langle h\pi(x)\Omega, \Omega \rangle, \quad h \in \pi(M)', \quad x \in M.$$

If $\omega \in M_*$ satisfies $|\omega(x)| \leq \psi(x)$ for all $x \in M^+$, then $\theta^{-1}(\omega)$ is uniquely defined and $\|\theta^{-1}(\omega)\| \leq 1$.

Proof sketch of (b). It is enough to prove $(\overline{F_* - M_*^+})^+ \subset F_*$.

Let $\omega \in (\overline{F_* - M_*^+})^+$ with sequences ω_n and φ_n such that $\omega_n \rightarrow \omega$ in norm of M_* and $\omega_n \leq \varphi_n \in F_*$ for all n . We may assume $\|\omega_n - \omega\| \leq 2^{-n}$ for all n by passing to a subsequence. Define

$$\psi := \omega + \sum_n [\omega_n - \omega] + \sum_n 2^{-n} \frac{\varphi_n}{1 + \|\varphi_n\|} \in M_*^+,$$

and let $\theta : \pi(M)' \rightarrow M_*$ be the commutant Radon-Nikodym map associated to ψ . Since $-\psi \leq \omega_n \leq \psi$ implies the boundedness $\|\theta^{-1}(\omega_n)\| \leq 1$ for all n , the weak convergence $\omega_n \rightarrow \omega$ in M_* implies the convergence $\theta^{-1}(\omega_n) \rightarrow \theta^{-1}(\omega)$ in the weak operator topology of $\pi(M)'$.

By the Mazur lemma, we can take a net ω_i in the convex hull of ω_n such that $\theta^{-1}(\omega_i) \rightarrow \theta^{-1}(\omega)$ strongly in $\pi(M)'$, and the corresponding $\varphi_i \in F^*$ can be defined such that $\omega_i \leq \varphi_i$ for all i . (In fact, the net ω_i can be taken to be a sequence because the commutant is σ -finite by the existence of the separating vector, but it is not necessary in here.) For each i and $0 < \delta < 1$, define

$$\omega_\delta := \theta(f_\delta(\theta^{-1}(\omega))), \quad \omega_{i,\delta} := \theta(f_\delta(\theta^{-1}(\omega_i))), \quad \varphi_{i,\delta} := \theta(f_\delta(\theta^{-1}(\varphi_i))).$$

Then, the strong convergence $f_\delta(\theta^{-1}(\omega_i)) \rightarrow f_\delta(\theta^{-1}(\omega))$ in $\pi(M)'$ implies $\omega_{i,\delta} \rightarrow \omega_\delta$ weakly in M_* , and the strong convergence $f_\delta(\theta^{-1}(\omega)) \rightarrow \theta^{-1}(\omega)$ in $\pi(M)'$ implies $\omega_\delta \rightarrow \omega$ weakly in M_* as $\delta \rightarrow 0$. If we define $\varphi_\delta := \theta(\lim_i f_\delta(\theta^{-1}(\varphi_i)))$ by taking a subnet or a cofinal ultrafilter, then $\varphi_{i,\delta} \rightarrow \varphi_\delta$ weakly in M_* . Since $\omega_i \leq \varphi_i$ and $0 \leq \varphi_{i,\delta} \leq \varphi_i \in F_*$, we get

$$\omega_{i,\delta} \leq \varphi_{i,\delta} \in F_*, \quad 0 \leq \omega_\delta \leq \varphi_\delta \in F_*, \quad \omega_\delta \in F_*, \quad \omega \in F_*. \quad \square$$

3.3 Strategies for (d)

Let $\omega \in (\overline{F^* - A^{*+}})^+$ with nets ω_i and φ_i such that $\omega_i \rightarrow \omega$ weakly* in A^* and $\omega_i \leq \varphi_i \in F^*$ for all i .

Question 1. How can we choose ψ ?

To construct ψ such that ω_i and φ_i are all absolutely continuous, it seems that we need to assume A is separable so that the weak* topology on the unit ball of A^* is metrizable and we may assume ω_i and φ_i are sequences. But even if this is the case, we can have $[\omega_n - \omega] \not\rightarrow 0$ in the weak* topology, contrary to the norm. So it seems difficult to construct ψ such that $\theta^{-1}(\omega_n)$ is bounded. However, we can still have $-2^n \leq \theta^{-1}(\omega_n) \leq \theta^{-1}(\varphi_n) \leq 2^n$, etc.

In fact, we should not have $\theta^{-1}(\omega_n)$ bounded. This is because the weak operator topology of $\pi(A)'$ is too strong. The topologies on $\pi(A)'$ generated by vectors $\pi(A)\Omega$ and $\pi^{**}(A^{**})\Omega$ are totally different when they are induced in A^* via θ , i.e. weak* and weak topologies. The weak (operator) convergence of $\theta^{-1}(\omega_n)$ implies the weak convergence of ω_n . The Mazur lemma tells us that we may assume $\omega_n \rightarrow \omega$ in norm. Something is strange.

Then, we can try to use the weak* compactness of the unit ball of A^* directly! Since $\theta^{-1}(\omega_n)$ is unbounded but controllable, we can try to take δ dynamically. We want to have the following properties:

$$-\delta_n < \theta^{-1}(\omega_n), \quad \theta(f_{\delta_n}(\theta^{-1}(\varphi_n))) \lesssim 1.$$

However, it is desparate to control the norm of $\theta(k)$ in terms of k , except by the norm of k . Also, by the “stiffness” or “rigidness” of the operator monotone functional calculi, two conditions are hardly satisfied simultaneously for non-commutative cases. When A is non-commutative, for example, $\omega_n \leq \varphi_n$ and $0 \leq \varphi_n$ do not imply $\omega_{n+} \leq \varphi_n$ because the ramp function is way far from operator monotone functions.

Solution 1. Take ψ dynamically!

Question 2. How can we make the weak* limit of ω_i commute with f_δ without strong topology?

Solution 2. We can approximate f_δ with affine functions by

$$t - \delta^{\frac{1}{2}} \leq f_\delta(t) \leq t, \quad |t| \leq 2^{-1}\delta^{-\frac{1}{4}}, \quad (1 + \delta^{-1})t \leq f_\delta(t) \leq t, \quad 0 \leq t \leq 1.$$

3.4 Proof of (d)

Proof of (d). As above, we will prove $(\overline{F^* - A^{*+}})^+ \subset F^*$. Define

$$G^* := \left\{ \omega \in A^{*sa} : \begin{array}{l} \text{there is } \psi \in A^{*+}, \text{ and there is } \varphi_\delta \in F^* \\ \text{for any sufficiently small } \delta > 0, \text{ such that} \\ \|\psi\| \leq 1, \|\varphi_\delta\| \leq \delta^{-1}, \text{ and } \omega \leq \varphi_\delta + \delta^{\frac{1}{2}}\psi \end{array} \right\}.$$

It suffices to show $F^* - A^{*+} \subset G^*$, $G^{*+} \subset F^*$, and $\overline{G^*} \subset G^*$.

Let $\omega \in F^* - A^{*+}$. Take $\varphi \in F^*$ such that $\omega \leq \varphi$. Define, for $\delta > 0$,

$$\psi := \frac{[\omega]}{1 + \|\omega\|} + \frac{\varphi}{(1 + \|\omega\|)(1 + \|\varphi\|)}, \quad \varphi_\delta := \theta(f_\delta(\theta^{-1}(\varphi))),$$

where θ is the commutant Radon-Nikodym map associated to ψ . The norm conditions $\|\psi\| \leq 1$ and $\|\varphi_\delta\| \leq \delta^{-1}$ are easily checked. For sufficiently small $\delta > 0$ such that $\|\theta^{-1}(\omega)\| \leq 1 + \|\omega\| \leq 2^{-1}\delta^{-\frac{1}{4}}$ and $\delta \leq 1$, we have

$$\theta^{-1}(\omega) \leq f_\delta(\theta^{-1}(\omega)) + \delta^{\frac{1}{2}} \leq f_\delta(\theta^{-1}(\varphi)) + \delta^{\frac{1}{2}},$$

so $\omega \leq \varphi_\delta + \delta^{\frac{1}{2}}\psi$ and $\omega \in G^*$.

Let $\omega \in G^{*+}$. Take $\psi \in A^{*+}$ and $\varphi_\delta \in F^*$ such that $\|\psi\| \leq 1$, $\|\varphi_\delta\| \leq \delta^{-1}$, and $\omega \leq \varphi_\delta + \delta^{\frac{1}{2}}\psi$, for any sufficiently small $\delta > 0$. Let $\psi_\delta := \omega + \delta\varphi + \psi$, and let θ_δ be the associated commutant Radon-Nikodym map. For any fixed $\delta' > 0$, since $0 \leq \theta_\delta^{-1}(\omega) \leq 1$, we have

$$0 \leq (1 + \delta')^{-1}\theta_\delta^{-1}(\omega) \leq f_{\delta'}(\theta_\delta^{-1}(\omega)) \leq f_{\delta'}(\theta_\delta^{-1}(\varphi_\delta + \delta^{\frac{1}{2}}\psi)) \leq f_{\delta'}(\theta_\delta^{-1}(\varphi_\delta) + \delta^{\frac{1}{2}}) \leq f_{\delta'}(\theta_\delta^{-1}(\varphi_\delta)) + \delta^{\frac{1}{2}},$$

and it implies $0 \leq (1 + \delta')^{-1}\omega \leq \theta_\delta(f_{\delta'}(\theta_\delta^{-1}(\varphi_\delta))) + \delta^{\frac{1}{2}}\psi_\delta$. Since $\|\psi_\delta\| \leq \|\omega\| + 2$ is bounded and $\theta_\delta(f_{\delta'}(\theta_\delta^{-1}(\varphi_\delta))) \in F^*$ is also bounded for fixed δ' as $\delta \rightarrow 0$, by considering the limit along a cofinal ultrafilter on the set of δ , we have $(1 + \delta')^{-1}\omega \in F^*$, so $\delta' \rightarrow 0$ gives $\omega \in F^*$.

To show G^* is weakly* closed, we claim for any $r > 0$ that

$$\overline{(F^* - A^{*+}) \cap A_{2r}^*} \subset G^*, \quad G^* \cap A_r^* \subset \overline{(F^* - A^{*+}) \cap A_{2r}^*},$$

where $A_r^* := \{\omega \in A^* : \|\omega\| \leq r\}$. If these are true, then

$$G^* \cap A_r^* = \overline{(F^* - A^{*+}) \cap A_{2r}^* \cap A_r^*}$$

is weakly* closed and convex in A^* for all $r > 0$, so the Krein-Šmulian theorem shows the claim.

Let $\omega_i \in (F^* - A^{*+}) \cap A_{2r}^*$ be a net such that $\omega_i \rightarrow \omega$ weakly* in A^* with $\|\omega_i\| \leq 2r$ for all i . Following the proof of $F^* - A^{*+} \subset G^*$, we can take $\psi_i \in A^{*+}$ and $\varphi_{i,\delta} \in F^*$ such that $\|\psi_i\| \leq 1$, $\|\varphi_{i,\delta}\| \leq \delta^{-1}$, $\omega_i \leq \varphi_{i,\delta} + \delta^{\frac{1}{2}}\psi_i$, for uniformly sufficiently small δ such that $1 + 2r \leq 2^{-1}\delta^{-\frac{1}{4}}$ because $\|\omega_i\|$ is bounded by $2r$. Since the three conditions are preserved by the weak* convergence, taking the limit along a cofinal ultrafilter on the index set of i , we can obtain limit points ψ and φ_δ so that $\omega \in G^*$.

Let $\omega \in G^* \cap A_r^*$. Take $\psi \in A^{*+}$ and $\varphi_\delta \in F^*$ such that $\|\psi\| \leq 1$, $\|\varphi_\delta\| \leq \delta^{-1}$, and $\omega \leq \varphi_\delta + \delta^{\frac{1}{2}}\psi$, for any sufficiently small $\delta > 0$. If $\delta^{\frac{1}{2}} < r$, then $\omega - \delta^{\frac{1}{2}}\psi \in (F^* - A^{*+}) \cap A_{2r}^*$ converges to ω weakly* in A^* as $\delta \rightarrow 0$, we have $\omega \in \overline{(F^* - A^{*+}) \cap A_{2r}^*}$. This completes the proof. $\square$